

Time Allowed: 3:00 hrs

Read the questions given below carefully and give your answers on the separate sheets provided

- Your explanations should be **brief and to the point**
- Write **neatly and legibly**

Part 1: Short answer Questions (20%)

1. What is the difference between abstract classes and interfaces?
2. What is the purpose/advantage of using interfaces?
3. What is the difference between checked and unchecked exceptions?
4. What is an event? Explain with an example in the context of graphical user interfaces?

unchecked - programmer can ignore it. It is b/c of error in a code
Abstract - no implementation
Interface - make no implementation
multiple
event
Abstract - no body

Part 2: Output Prediction(30%)

```
public abstract class Student {  
    private String fullName;  
    private String idNumber;  
    protected double registrationFee;  
    public Student(String name, String id){  
        fullName = name;  
        idNumber = id;  
    }  
    public void payRegistrationFee(){  
        registrationFee = 50.0;  
        System.out.println("Registration fee:"  
            + registrationFee );  
    }  
    public abstract void payTution();  
}
```

INSTRUCTION Output Prediction:

Consider the following four classes given and **determine the output of the code inside the main method of the StudentTest class.**

shapes[0] = new Shapes
Students[0]

6 10
3 to 10
21

```
public class RegularStudent extends Student {  
    private double accomdationCost, foodCost, educationCost;  
  
    public RegularStudent(String name, String id, double a, double f, double e){  
        super(name, id);  
        accomdationCost = a;  
        foodCost = f;  
        educationCost = e;  
    }  
  
    public void payTution(){  
        double tution = accomdationCost + foodCost + educationCost;  
        System.out.println("Tution fee for regular student is " + tution);  
    }  
}
```

@ override
public class speed calculator (C) {

```

public class ExtensionStudent extends Student{

    private int creditHr;
    private double feePerCreditHr;

    public ExtensionStudent(String name, String id, int hr, double fee){
        super(name,id);
        creditHr = hr;
        feePerCreditHr = fee;
    }

    public double getCreditHr(){return creditHr;}
    public void setCreditHr(int hours){creditHr = hours;}

    public void payTution(){
        double tution = creditHr * feePerCreditHr;
        System.out.println("Tution fee for extension student is " + tution);
    }

    public void payRegistrationFee(){
        if (creditHr < 50)
            super.payRegistrationFee();
        else{
            registrationFee = 100;
            System.out.println(" The fee for registering this student is " +
registrationFee );
        }
    }
}

```

```

public class StudentTest {
    public static void main(String[] args) {

        Student[] students = new Student[2];

        students[0] = new RegularStudent("Abebe", "ETR/0001/00", 1000, 2000, 3000);
        students[1] = new ExtensionStudent("Kebede", "ETR/0002/00", 75, 100.00);

        for(int i = 0; i < students.length; i++){
            System.out.println("Calculating Tution for Student" + (i+1));
            students[i].payRegistrationFee();
            students[i].payTution();
            System.out.println();
        }
        RegularStudent students3 = (RegularStudent)students[0];
        ExtensionStudent students4 = (ExtensionStudent)students[1];

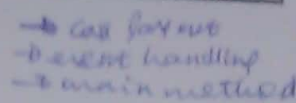
        students4.setCreditHr(45);
        students4.payRegistrationFee();
        students4.payTution();
    }
}

```

output

Students 1	Tutution = 6000
Students 2	= 7500
Students 3	= 6000
Student 4	= 3250

Students 3 $\Rightarrow$ Tution = 6000
 Students 4, Tution =



20-6

Component $\frac{1}{2} \text{ mol}$

CP. 1000

- Speed

flow layout

super(2)

1. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$